

Methodology 3

Cellular communication : Methods for studying protein interactions

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Course plan

I. Introduction

II. Definition and characterization of protein domains

III. Common methods for studying protein interactions

Course plan

I. Introduction

II. Definition and characterization of protein domains

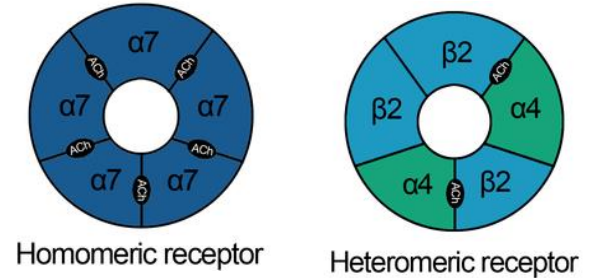
III. Common methods for studying protein interactions

I. Introduction

Protein-protein interactions (PPIs) handle a wide range of biological processes: Enzyme activity, cell adhesion, cell signaling etc...

PPIs can be classified in several ways, based on their:

- 1) Interaction surface: homo- or hetero-oligomers
- 2) stability: obligate or non-obligate



Obligate complex :

Subunits are misfolded and unstable when alone



Non-obligate complex

Subunits are well folded and stable when alone

- 3) persistence: transient or permanent

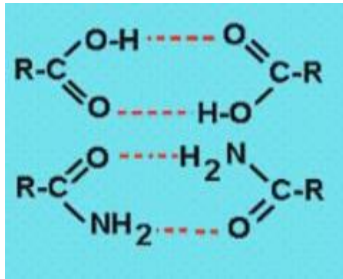
The study a protein's interacting partners is important to infer the protein's function(s) and to identify new drug targets.

I. Introduction: bond types

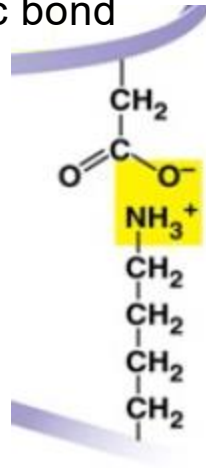
- Interaction between subunits or protein partners can be mediated by:

1. non-covalent weak bonds

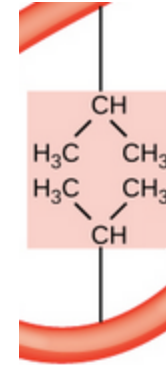
Hydrogen bond



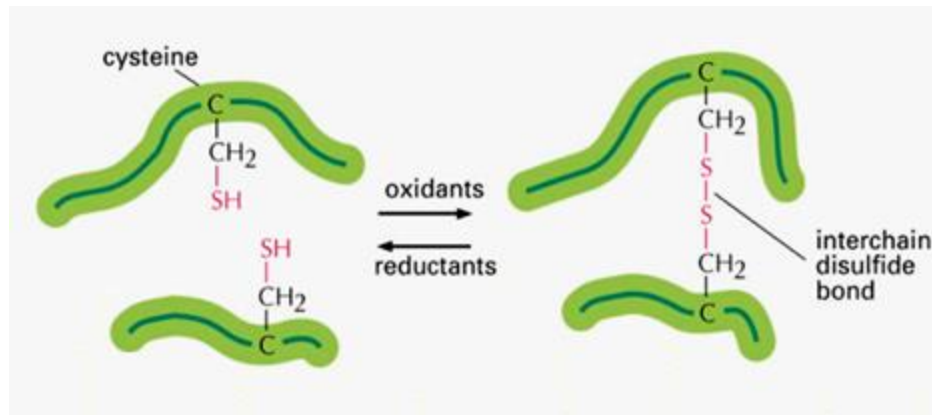
Ionic bond



Hydrophobic bond
(Van der Waals interaction)



2. covalent bonds: Disulfide bridges



To study/identify PPIs, these interactions must be preserved

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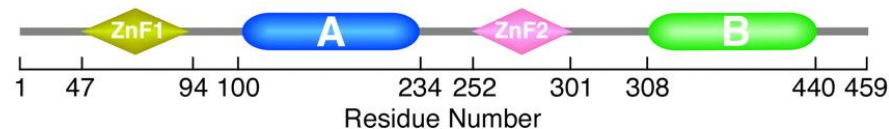
III. Common methods for studying protein interactions

What is a protein domain ?

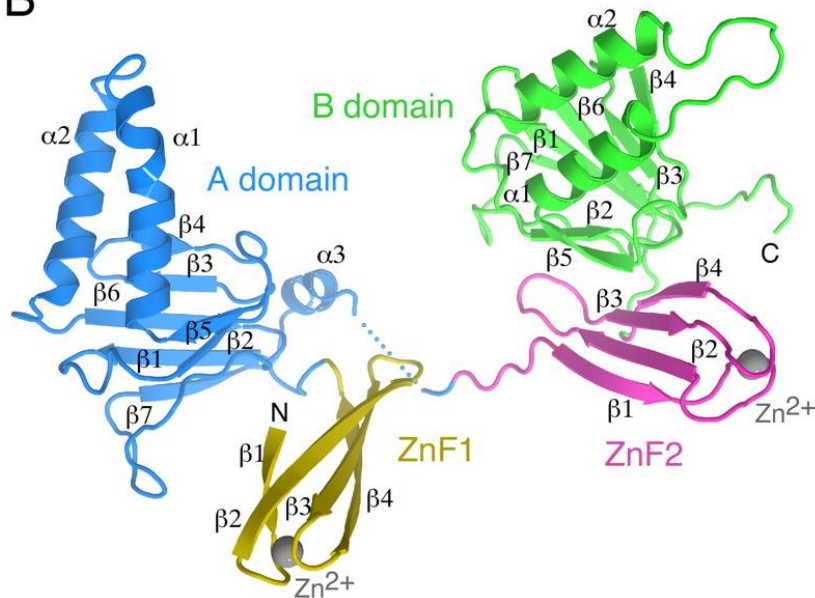
Proteins are constituted of protein domains. Certain protein domains are essential in PPI

A protein domain is a conserved part (module) of a given protein sequence (40 to 250 aa, with critical ones) and tertiary structure that can evolve, function, and exist independently of the rest of the protein

A



B



Example of the **conserved primary sequence** of the **SH2 domain** of the **Src kinase protein family**.

Src	human	61	SIQAEEWYFGKITRRESERLLLNAENPRGTFLVRESETTKGAYCLSVSDFDNAK
	mouse	50	SIQAEEWYFGKITRRESERLLLNAENPRGTFLVRESETTKGAYCLSVSDFDNAK
	chick	62	SIQAEEWYFGKITRRESERLLNPENPRGTFLVRESETTKGAYCLSVSDFDNAK
	<i>Drosophila</i>	156	SVNSEDWFFENVLRKEADKLLLAENPRGTFLVRPSEHNPNNGYSLSVKDWEDGR
Fyn	human	143	SIQAEEWYFGKLGRKDAERQLLSFGNPRGTFLIRESETTKGAYSLSIIRDWDDMK
	mouse	143	SIQAEEWYFGKLGRKDAERQLLSFGNPRGTFLIRESQTTKGAYSLSIIRDWDDMK
	chick	143	SIQAEEWYFGKLGRKDAERQLLSFGNPRGTFLIRESETTKGAYSLSIIRDWDDMK
Lck	human	121	SLEPEPWFFKNLSRKDAERQLLAPGNTHGSFLIRESESTAGSFSLSVRDFDQNO
Lyn	human	123	TLETEEWFFKDI TRKDAERQLLAPGNSAGAF LI RESE TLKGSFSLSVRDFDPVH
Src	human	115	GLNVKHYKIRKLDSSGGFYITSRTQFNSLQQLVAYYSKHADGLCHRLTTVCPTSK
	mouse	104	GLNVKHYKIRKLDSSGGFYITSRTQFNSLQQLVAYYSKHADGLCHRLTTVCPTSK
	chick	116	GLNVKHYKIRKLDSSGGFYITSRTQFSSLQQLVAYYSKHADGLCHRLTNVCPTSK
	<i>Drosophila</i>	210	GYHVKHYRIKPLDNGGYIATNQTFPSLQALVMAYSKNALGLCHILSRPCPKPQ
Fyn	human	197	GDHVKHYKIRKLDNGGYIITTRAQFETLQQLVQHYSEAAAGLCCRLVVP
	mouse	197	GDHVKHYKIRKLDNGGYIITTRAQFETLQQLVQHYSEKADGLCFNLTVV
	chick	197	GDHVKHYKIRKLDNGGYIITTRAQFETLQQLVQHYSEKADGLCFNLTVI
Lck	human	175	GEVVKHYKIRNLDNGGGFYISPRITFPGLHELVRHYTNASDGLCTRLSRPCQTQK
Lyn	human	177	GDVIKHYKIRSLDNGGYIISPRITFPCISDMIKHYQKQADGLCRRLEKACISPK

Figure 30 Partial amino acid sequence alignment of the SH2 domains from four members of the Src family of tyrosine kinases: Src, Fyn, Lck and Lyn. For Src and Fyn, amino acid sequence data for several different species are included. For each protein, numbering is based on the most N-terminal amino acid in the polypeptide being numbered 1. Amino acid identities are highlighted in green.

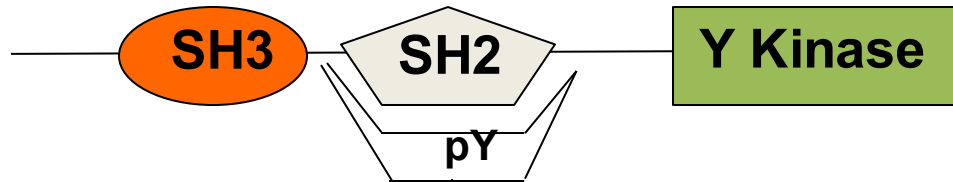
The primary sequence of SH2 domains from various proteins of the Src family are conserved. The sequence is also highly homologous between species.

Domains may carry certain specific functions of the complete protein: binding of ligands, interaction with other macromolecules, catalytic site...

Function	Protein domain
Interaction after post-translational modification	
Phosphotyrosine	SH2, PTB
Phosphoserine/Phosphothreonine	14-3-3, WD40, FHA
Interaction with protein motifs	
Proline rich motifs	SH3, WW, EVH1
Protein C-terminus	PDZ
Regulation of specific signaling pathways	
Proteins involved in GPCR signaling	GoLoco, RGS, RBD, GGL
Proteins regulating apoptosis	DD, DED, CARD, BH1-4
Proteins regulating ubiquitinylation	F-box, hect, RING, UBA
Calcium binding motifs	EF-hand, C2
Membrane interaction	
Phospholipids, DAG	PH, C2, FYVE, C1, PX, ENTH, tubby
DNA interaction	
Chromatin	Bromo or Chromo domain
DNA	Zinc finger, Leucine zipper, bHLH
Enzymatic activity	
Serine/ threonine kinase activity	S/T kinase domain
Phospholipid metabolism	Phospholipase D domaine
GMP cycling	Cyclase domain

Conserved function of an isolated domain

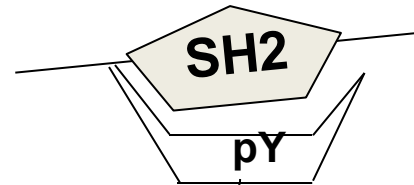
Src



FAK



SH2 domain of Src



FAK

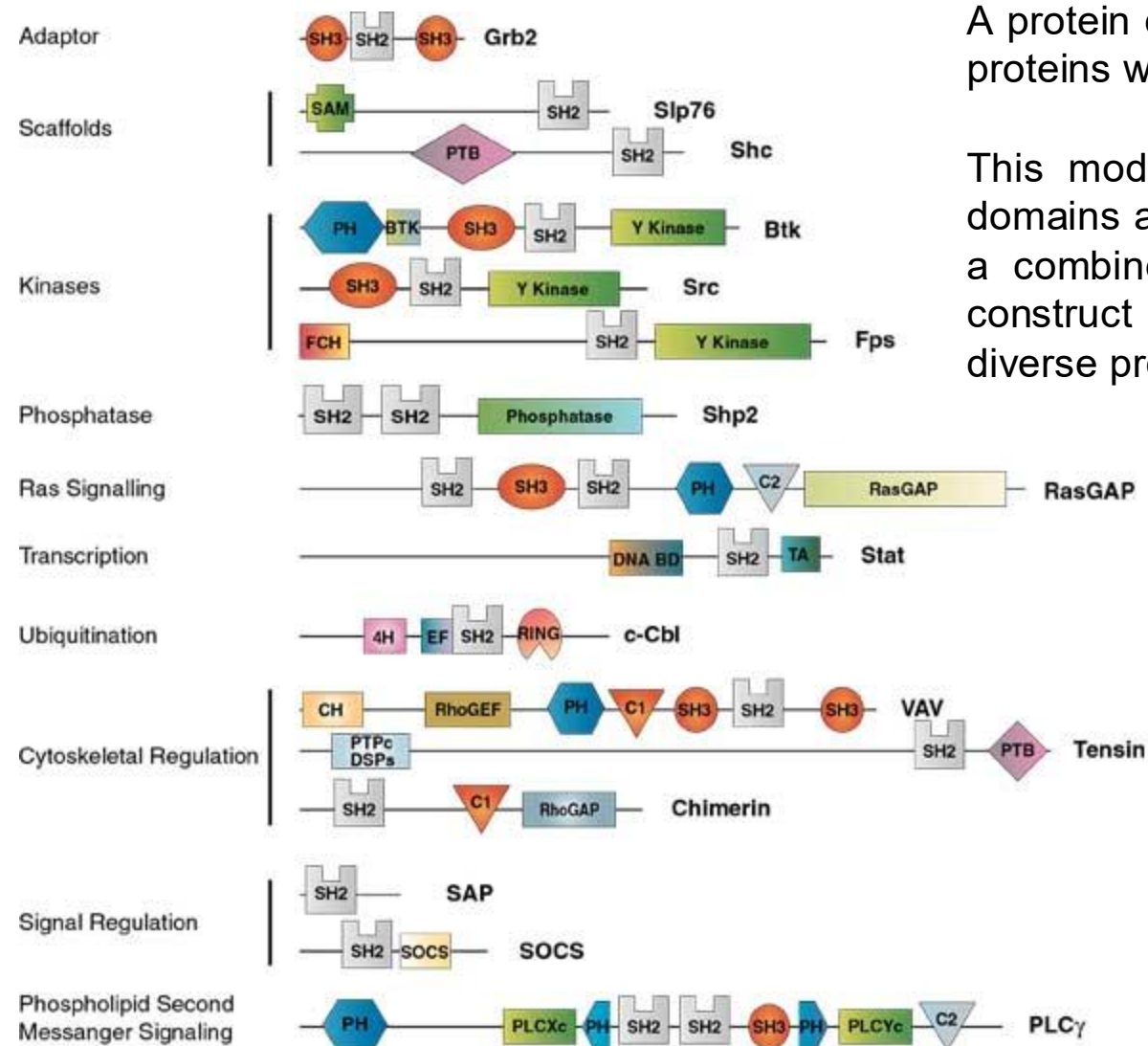


- A protein domain can be produced and isolated from its natural environment
- **It keeps most of its functional properties**

Proteins are composed of multiple domains

A protein domain may be present in different proteins with different functions.

This modular organization of proteins into domains allows these domains to be used in a combined manner as building blocks to construct a wide variety of functionally diverse proteins.



Course plan

I. Introduction

II. Definition and characterization of protein domains

III. Common methods for studying protein-protein interactions

Many approaches to study PPIs

Many methods for exploring PPIs exist:

In vitro methods are performed in a controlled environment outside a living organism.
For example co-immunoprecipitation, GST pull-down assays, surface plasmon resonance.

In vivo methods are performed in a living organism.
For example yeast two hybrid, FRET imaging

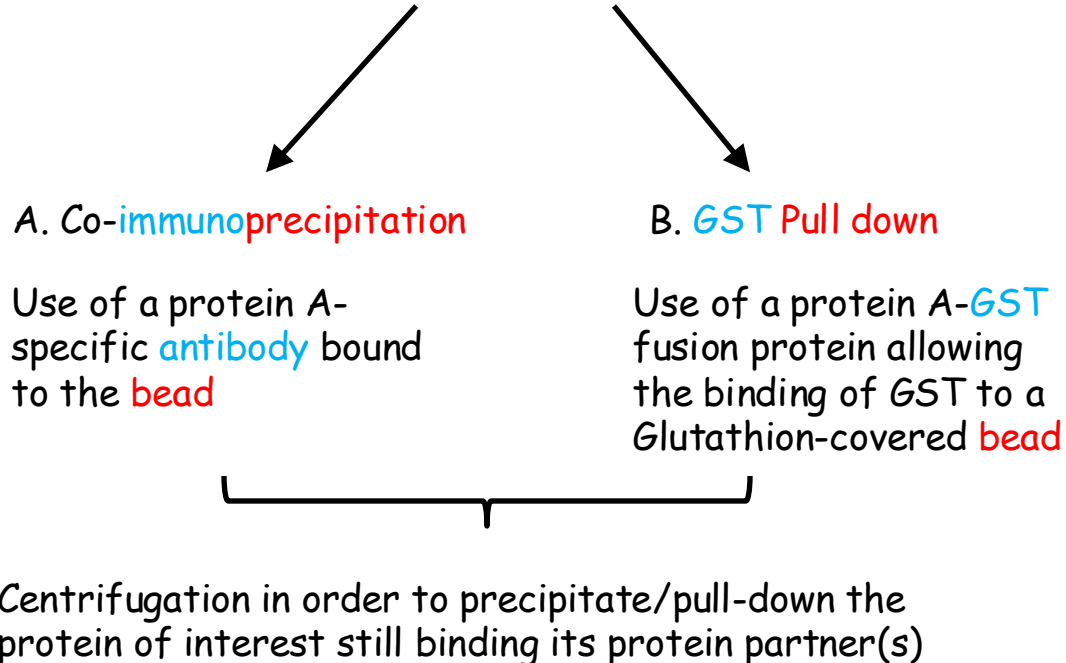
In silico methods are performed on a computer or via computer simulation.
For example: generation of phylogenetic trees

The methods used will be chosen according to the question asked:

Do I want to identify an interaction domain ? Do I want to identify new partners of my protein? Do I want to validate a suspected partner ? Do I want to study the regulation of the interaction ? Do I want to characterize kinetic properties of the interaction ? Do I want to localize the interaction within the cell ?....

(A+B) Analysis of protein interactions using precipitation / pull-down assays

Principle : Isolation of a protein of interest (protein A) together with its binding partner(s) from a large pool of proteins (a cell lysate or medium) with a **heavy bead** indirectly bound to the protein of interest in order to allow its isolation by **precipitation (=pull down)**, with the help of centrifugation.



A. Co-**immunoprecipitation**

Use of a protein A-specific **antibody** bound to the **bead**

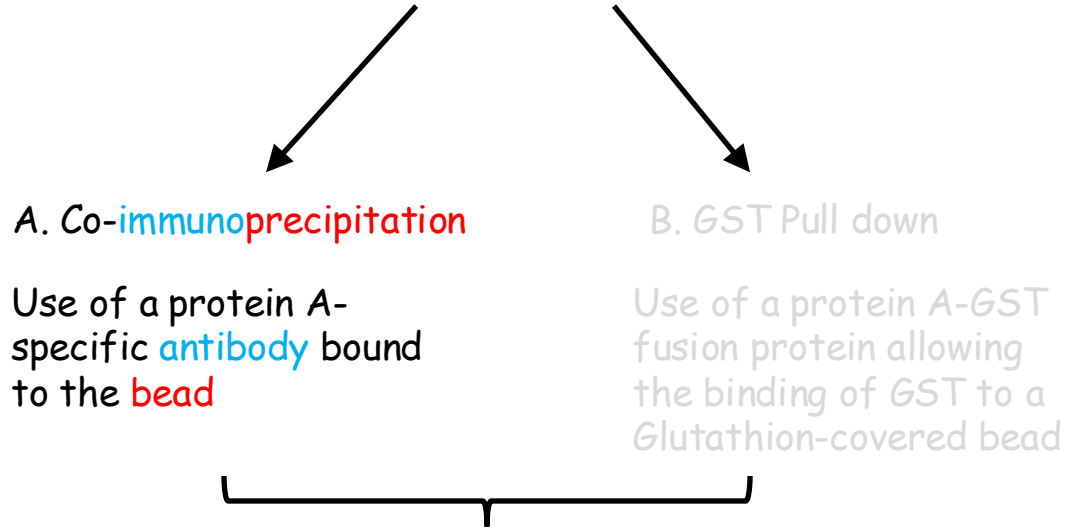
B. **GST** Pull down

Use of a protein A-**GST** fusion protein allowing the binding of GST to a Glutathion-covered **bead**

Centrifugation in order to precipitate/pull-down the protein of interest still binding its protein partner(s)

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A. Co-immunoprecipitation

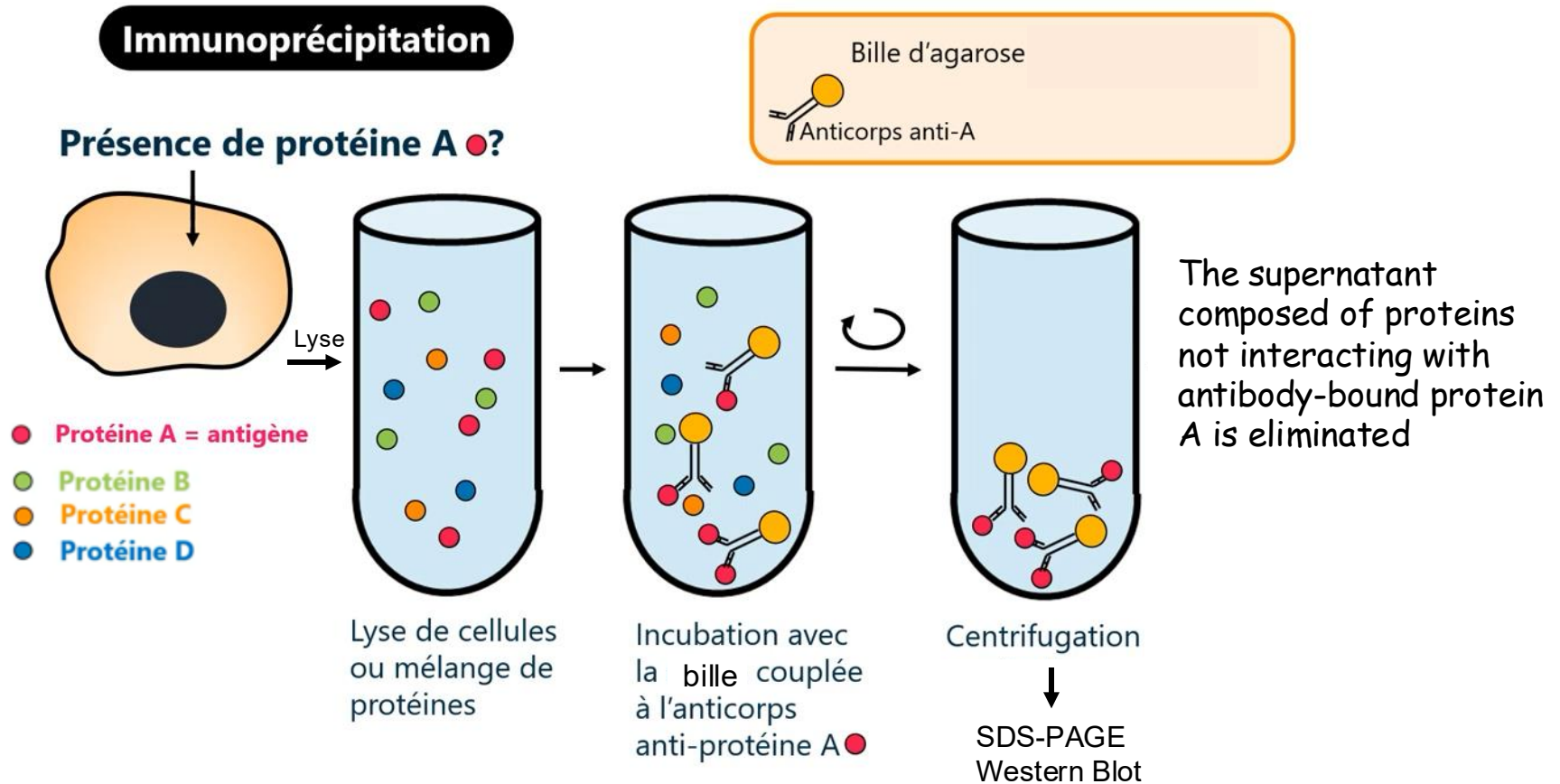
Use of a protein A-specific antibody bound to the bead

B. GST Pull down

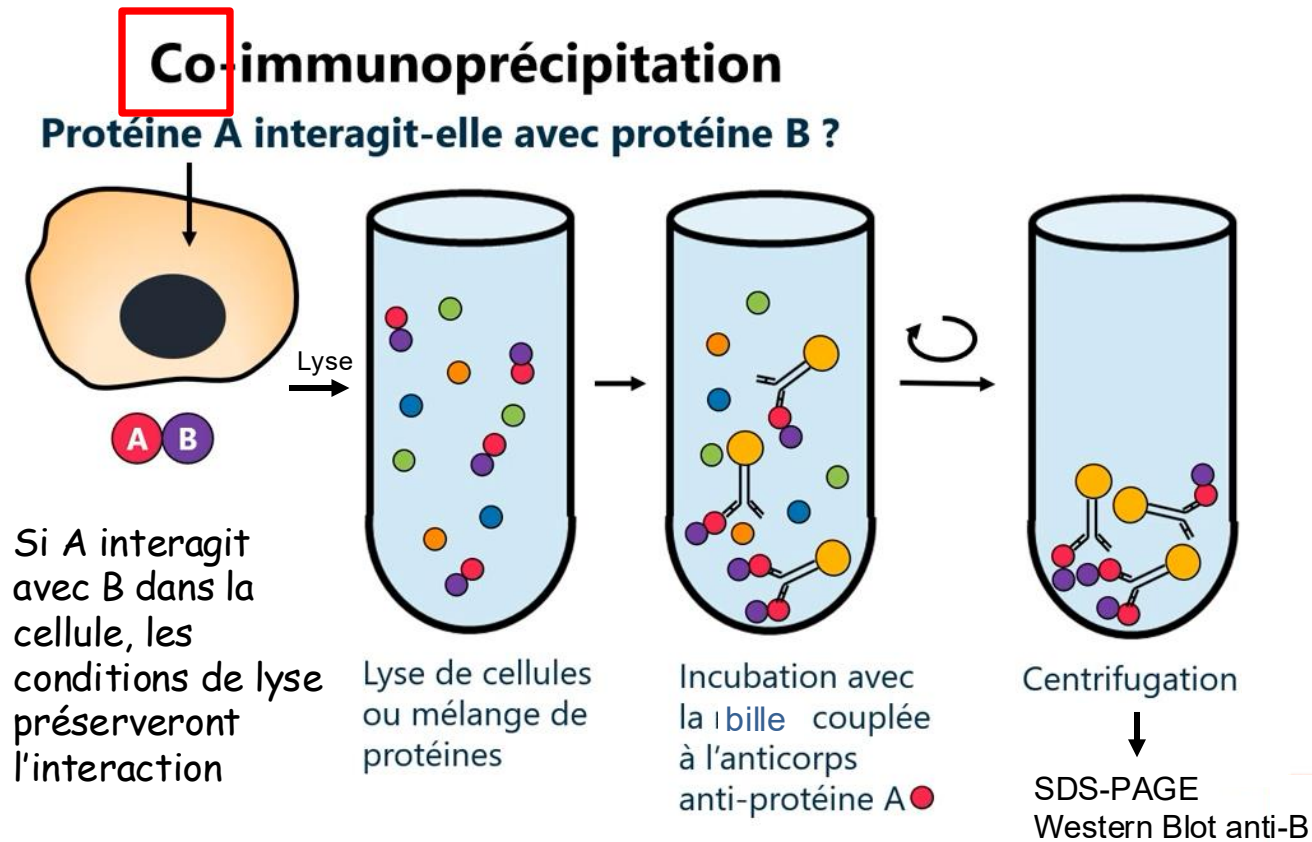
Use of a protein A-GST fusion protein allowing the binding of GST to a Glutathion-covered bead

Centrifugation in order to precipitate/pull-down the protein of interest still binding its protein partner(s)

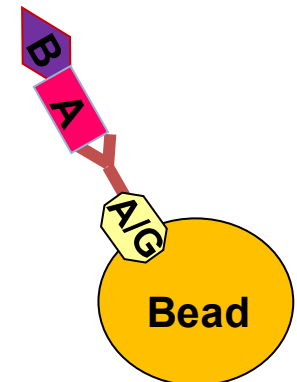
A. Analysis of protein interactions using co-immunoprecipitation



A. Analysis of protein interactions using co-immunoprecipitation



The antibody (directed against protein of interest) is coupled to protein A/G-coated beads (agarose, magnetic beads) (protein A/G=strong IgG Fc domain binding protein)



Note: protein A/G-coated magnetic beads can also be used. Beads will be precipitated with a magnet instead of centrifugation

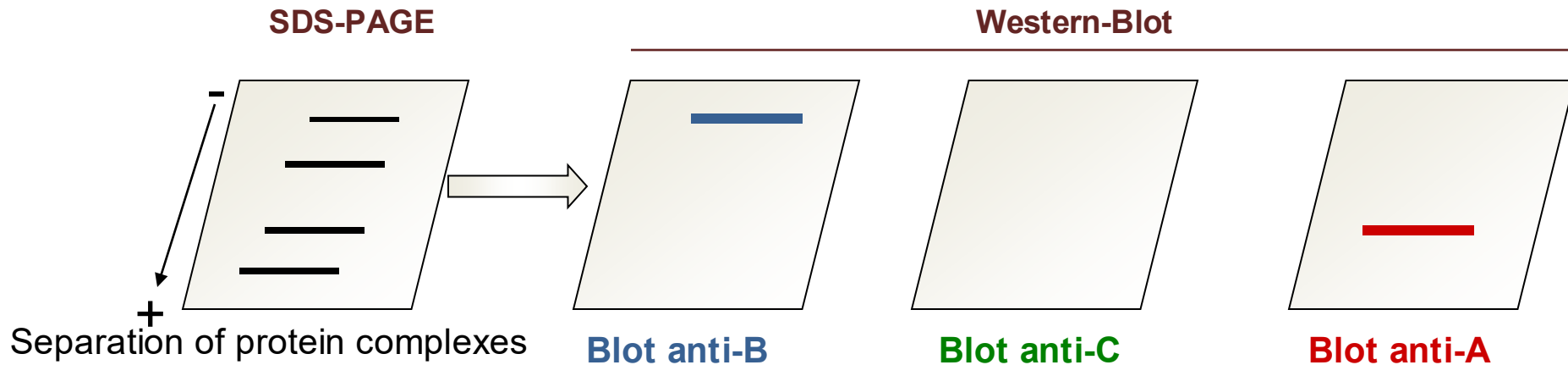
A. Analysis of protein interactions using co-immunoprecipitation

Identification of co-immunoprecipitated proteins

a) If Y's partners are known or suspected

b) If Y's partners are unknown

Western Blot with SDS-PAGE
(+/- denaturing conditions)
with anti-'suspected partner'
antibody

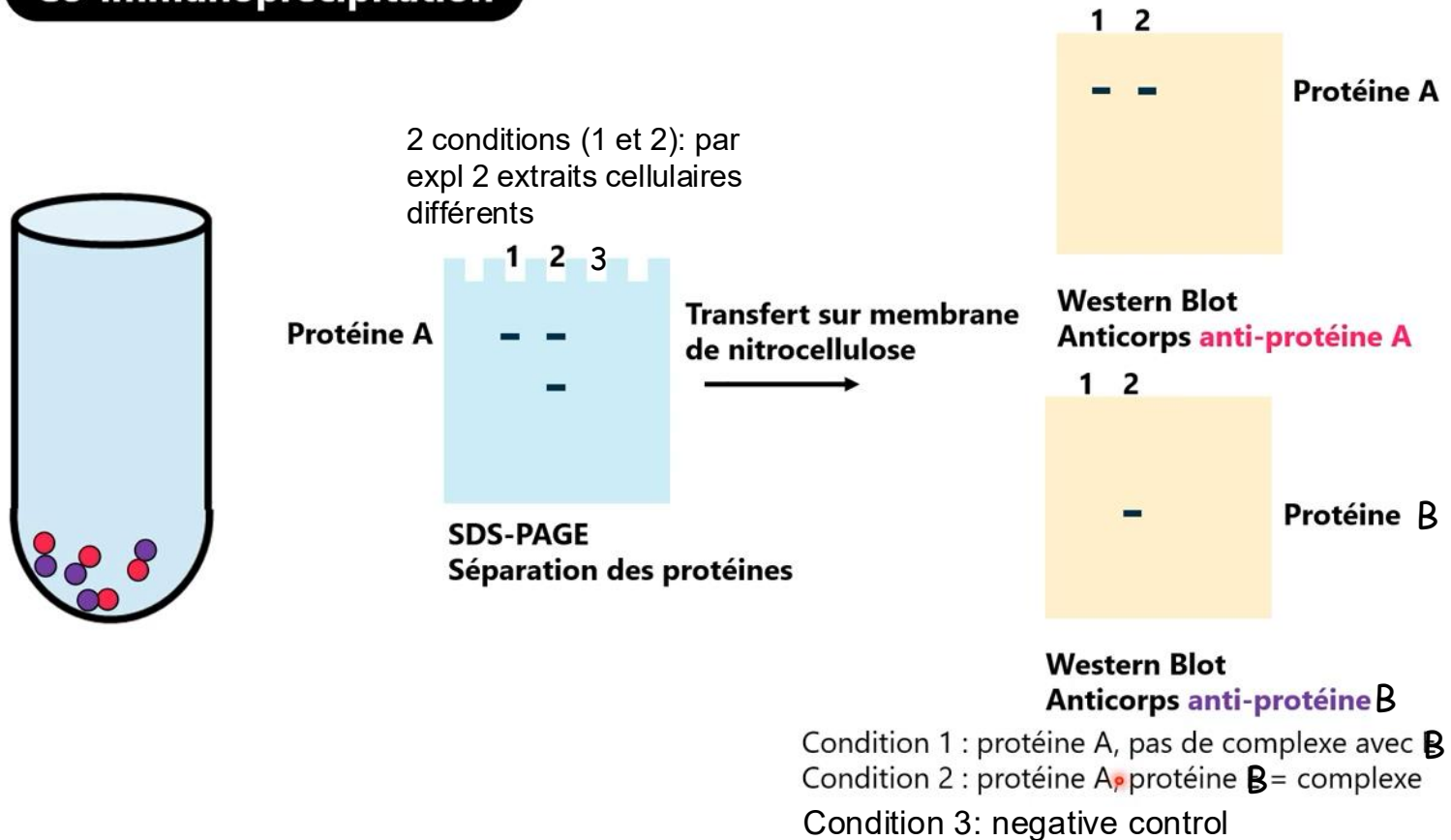


Conclusions : **A** and **B** are present in the protein complex
C is not present in the protein complex

Whether the binding of A and B is direct or via another protein cannot be concluded

A. Analysis of protein interactions using co-immunoprecipitation

Co-immunoprécipitation



Possible controls:

- Use of a pre-immune antibody to test potential non-(protein A-) specific binding
- When using transfected cells, use mock- (empty vector-) transfected cells as negative control and B-expressing cells as positive controls.
- Use tissues from gene B or A knock-out models.

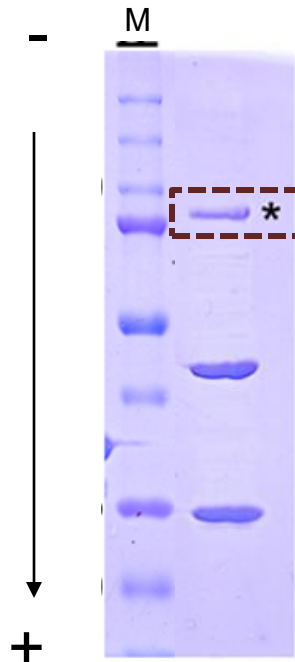
A. Analysis of protein interactions using co-immunoprecipitation

Identification of co-immunoprecipitated proteins

a) If Y's partners are known or suspected

b) If Y's partners are unknown

SDS-PAGE and Coomassie staining



Cut out the band, elution and **sequencing** of the protein
(Edman's and MALDI-TOF methods in appendix)

Comparison of the obtained peptide sequence with
protein sequences from databases



Identification of the protein

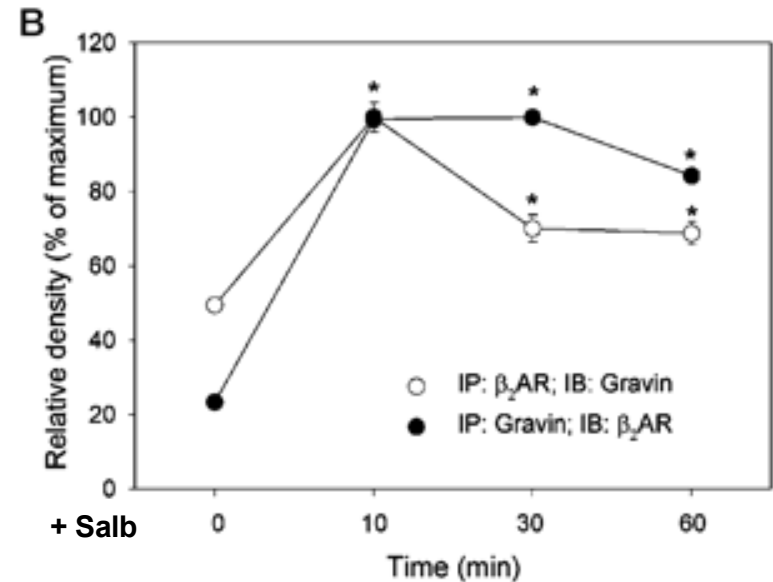
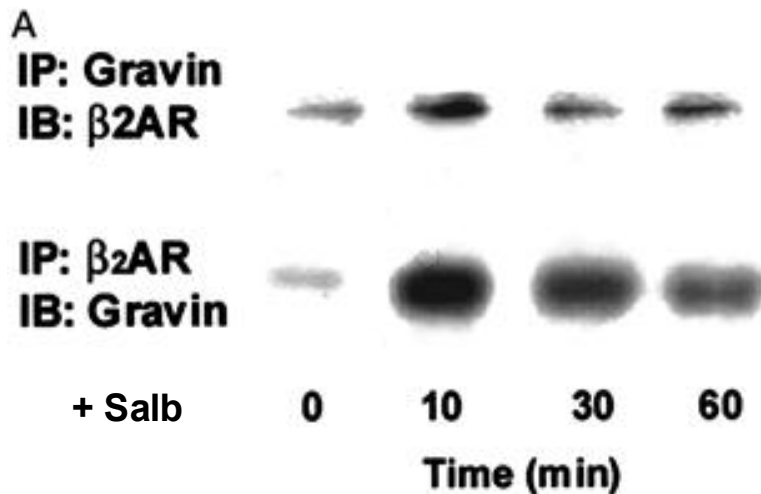
Separation of protein complexes

Or start directly from IP after elution of proteins (from antibodies) collected in the pellet

A. Analysis of protein interactions using co-immunoprecipitation

Example: Analysis of the regulated interaction between Gravin and the β_2 -adrenergic receptor

- Stimulation of HASM (human airway smooth muscle) with the receptor agonist Salbutamol (10 μ M) for 60 minutes
- Lysis of cells and (co-) IP
- Validation of the suspected interacting partner by immunoblotting (IB)



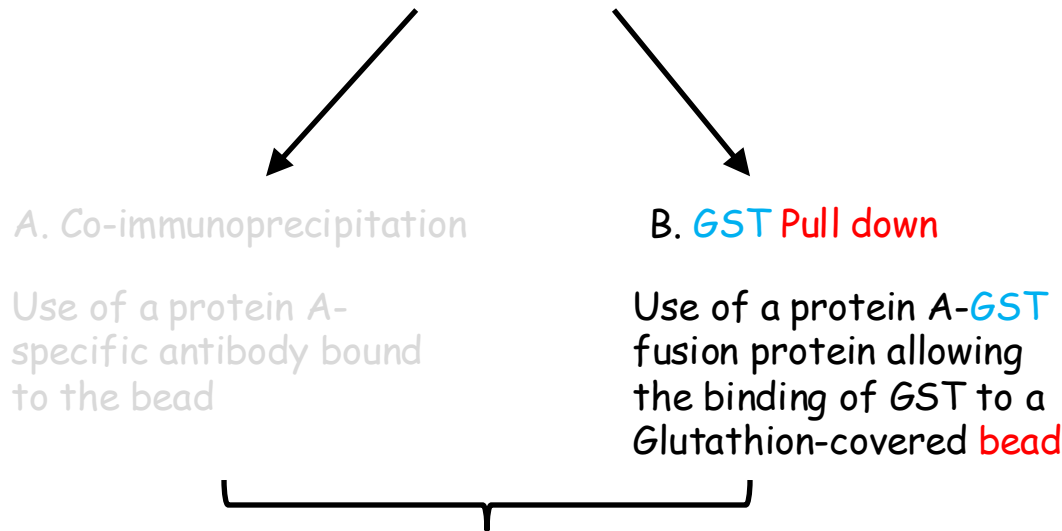
Q1. Analyze figures (goal, method, results and conclusions).

Q2. What controls are missing to validate the result ?

Q3: If antibodies for gravin and β_2 AR were not available, could you still do a co-IP for these proteins?

(A+B) Analysis of protein interactions using precipitation / pull-down assays

Principle : Isolation of a protein of interest (protein A) together with its binding partner(s) from a large pool of proteins (a cell lysate or medium) with a **heavy bead** indirectly bound to the protein of interest in order to allow its isolation by **precipitation (=pull down)**, with the help of centrifugation.



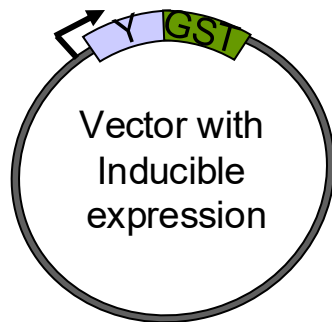
Centrifugation in order to precipitate/pull-down the protein of interest still binding its protein partner(s)

B. Analysis of protein interactions using GST-pull down

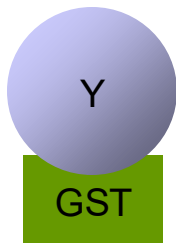
GST pull-down can be used to validate suspected interactions, identify new interaction partners, identify interaction domains, test direct interaction between purified proteins

Principle : Use a recombinant protein Y fused to GST (Glutathione S-Transferase) to pull down its potential partners.

1) Production and purification of recombinant protein Y fused to GST

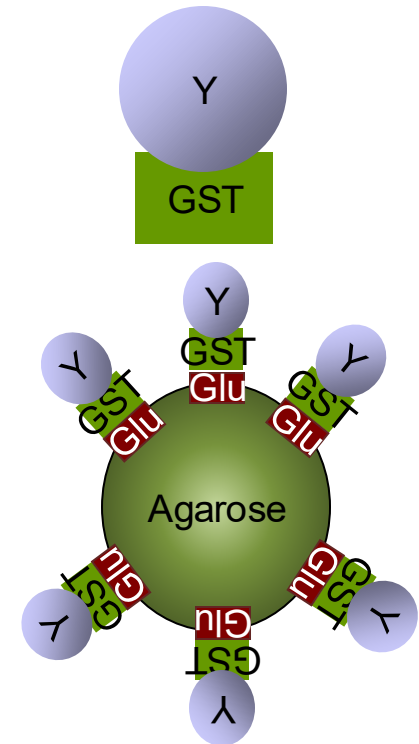


Transfection, Selection,
Protein expression and Purification



Binding of Y-GST to glutathione-coupled
agarose beads
(due to affinity of GST for glutathion)

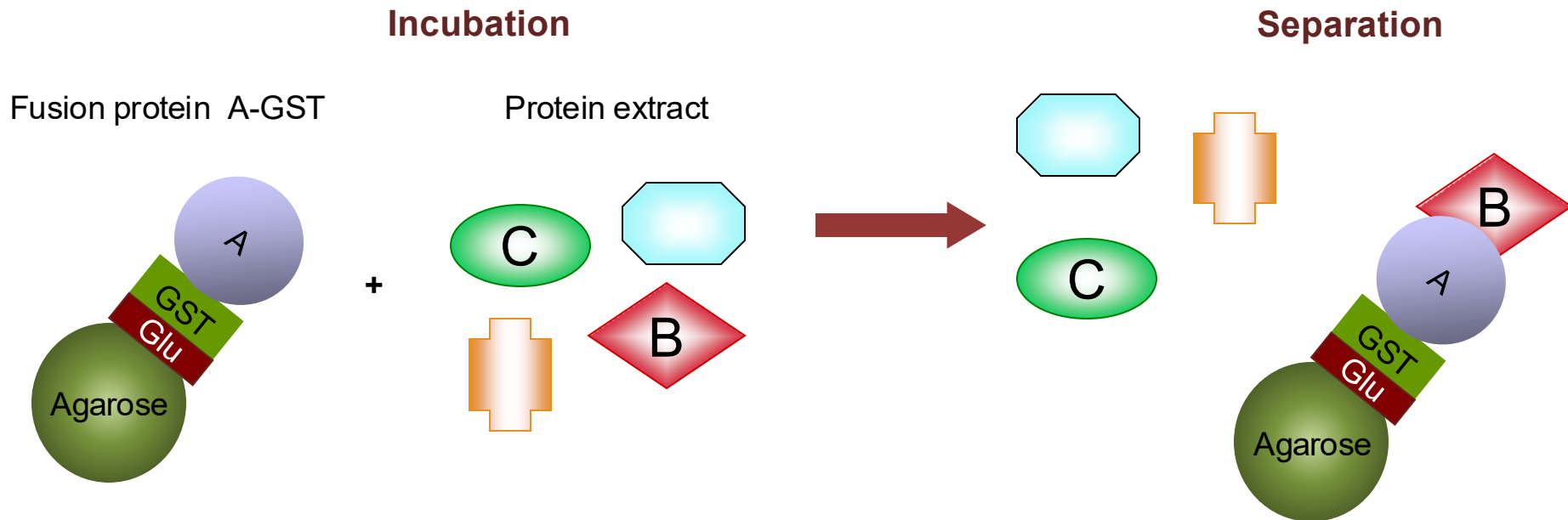
Purified fusion protein Y-GST



B. Analysis of protein interactions using GST-pull down

2) **Incubation of the protein extract** (plasma, cell lysate, conditioned medium, purified proteins ...)

3) **Separation of bound** (to A-GST complexed protein / agarose bead) and unbound proteins by precipitation of agarose beads (centrifugation)

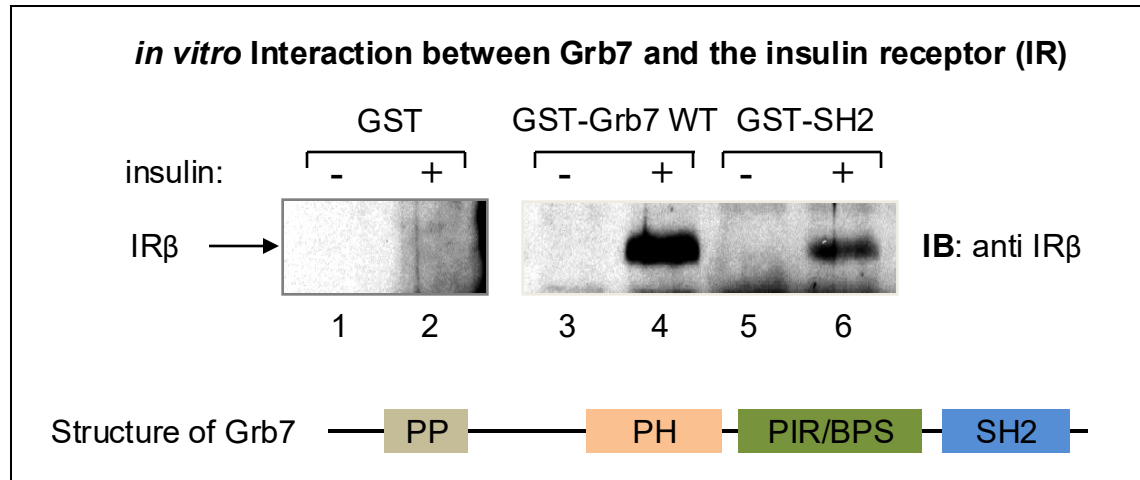


- Use purified non-fused GST as a negative control to detect non specific binding
- When using transfected cells, use mock transfected cells as negative control and Y expressing cells as positive controls.
- Use tissues from from gene Y or X knock-out models as negative controls if possible.

B. Analysis of protein interactions using GST-pull down

4) Identification of binding partners

a) If Y's partners are known or suspected → Western Blot after SDS-PAGE



b) If Y's partners are unknown → Purification and sequencing of the protein from SDS-PAGE band

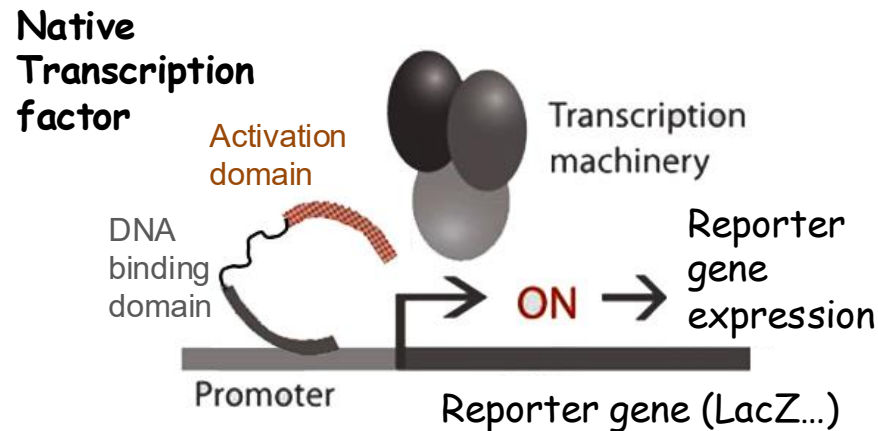
Q1: Analyze the figure.

Q2: Why do researchers use GST alone ?

C. Analysis of protein interactions using Yeast two hybrid

The 'two (/double) hybrid' technique is a molecular biology technique for detecting an **interaction** between two proteins.

Principle : Yeast two-hybrid is based on the reconstitution of a functional transcription factor when two proteins of interest interact



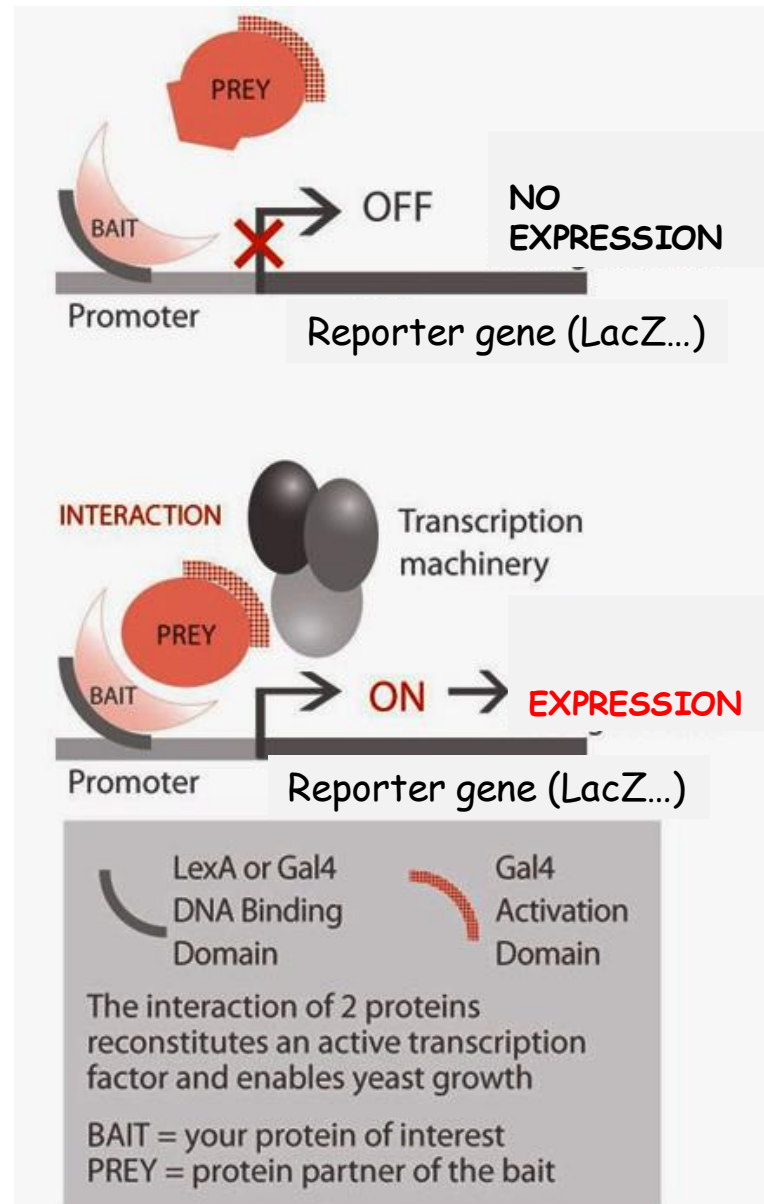
C. Analysis of protein interactions using Yeast two hybrid

The transcription factor has been cut in two parts.

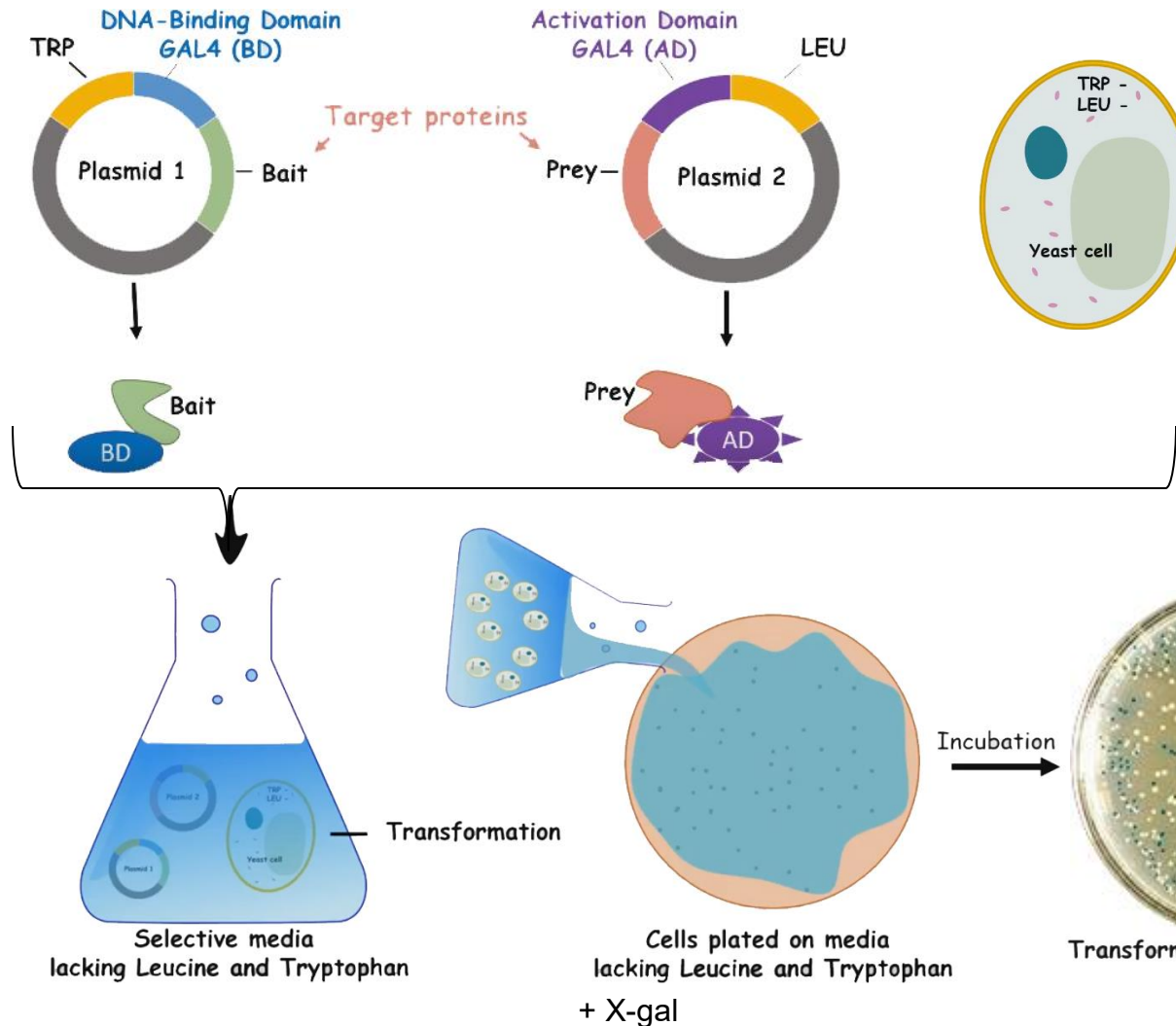
The protein of interest, called **bait**, is fused to the DNA binding domain of the transcription factor (=hybrid 1)

The suspected partner of the bait, called **prey**, is fused to the activation domain of the transcription factor (= hybrid 2)

When both hybrids interact, the transcription factor is functionally reconstituted and will drive the transcription of a reporter gene



C. Analysis of protein interactions using Yeast two hybrid



In plasmid 2 a whole library of cDNAs can be subcloned and tested (when there is no specific suspect)

Only yeast transformed with both plasmids will synthesize Leucine + Tryptophan and thus survive

Yeast cells are also expressing a reporter gene (Lac Z...) whose expression is driven by a promoter activated by the reconstituted transcription factor

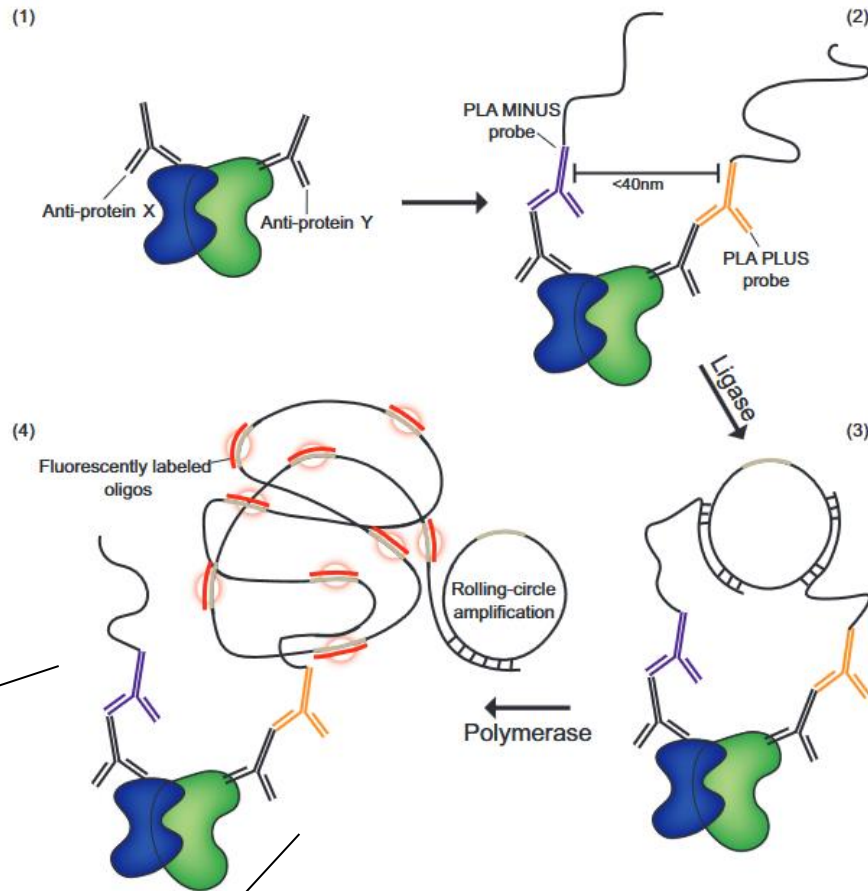
Sequencing of positive clones (=blue clones, if lacZ is the reporter gene) allowing identification of the preys (=partners)

LacZ produces β -galactosidase, which can be detected using colorimetric substrates like X-gal (transformed into a blue product by β -gal).

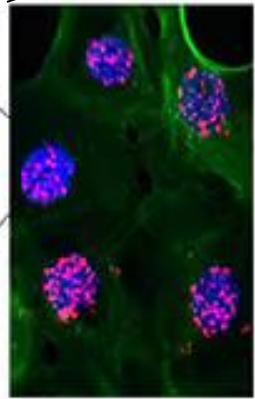
Other reporter or selection genes can be used (HIS3 or ADE2: Allow growth on media lacking histidine or adenine).

D. Analysis of protein interactions using Proximity Ligation Assay

Proximity Ligation Assay (PLA) is a powerful technique which allows protein interactions to be detected in situ.

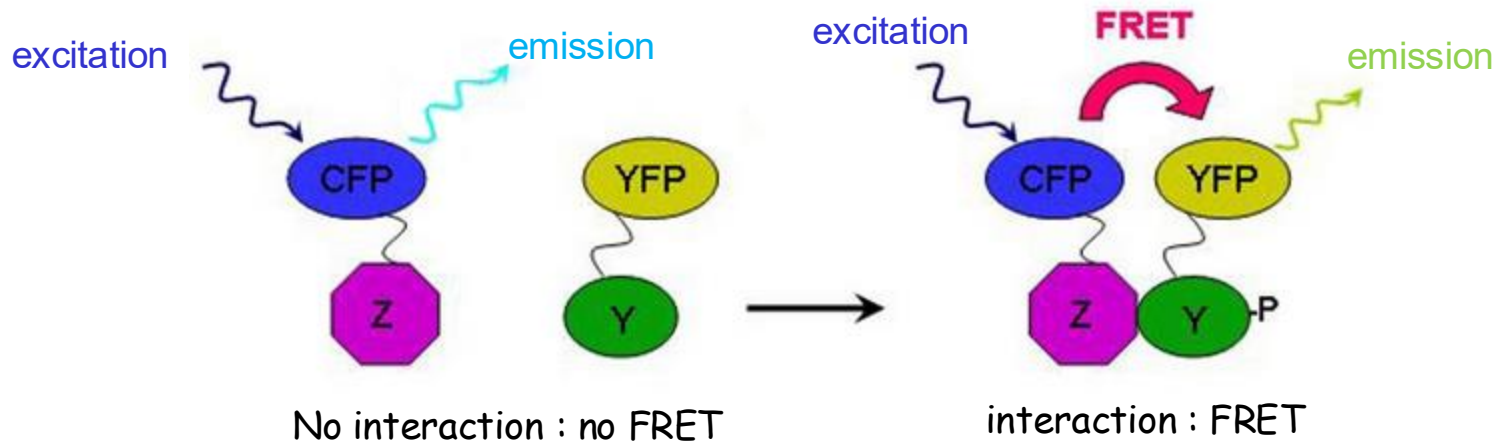


(1) Two antibodies of different species are selected that recognize two proteins of interest. **(2)** Secondary antibodies that recognize the Fc regions of the primary antibody species are conjugated with oligonucleotide probes (PLA probes) and then incubated with the sample. **(3)** Ligase and free oligonucleotides that hybridize with the probes bring the oligonucleotides together in a closed circle conformation if the proteins are in close enough proximity. **(4)** Finally, the circular and closed DNA matrix thus formed can be amplified by the addition of polymerase, and fluorescently labeled oligonucleotides can then hybridize with repeated sequences within the amplicons resulting in the accumulation of a fluorescent spot in areas of successful amplification and thus of protein interactions.

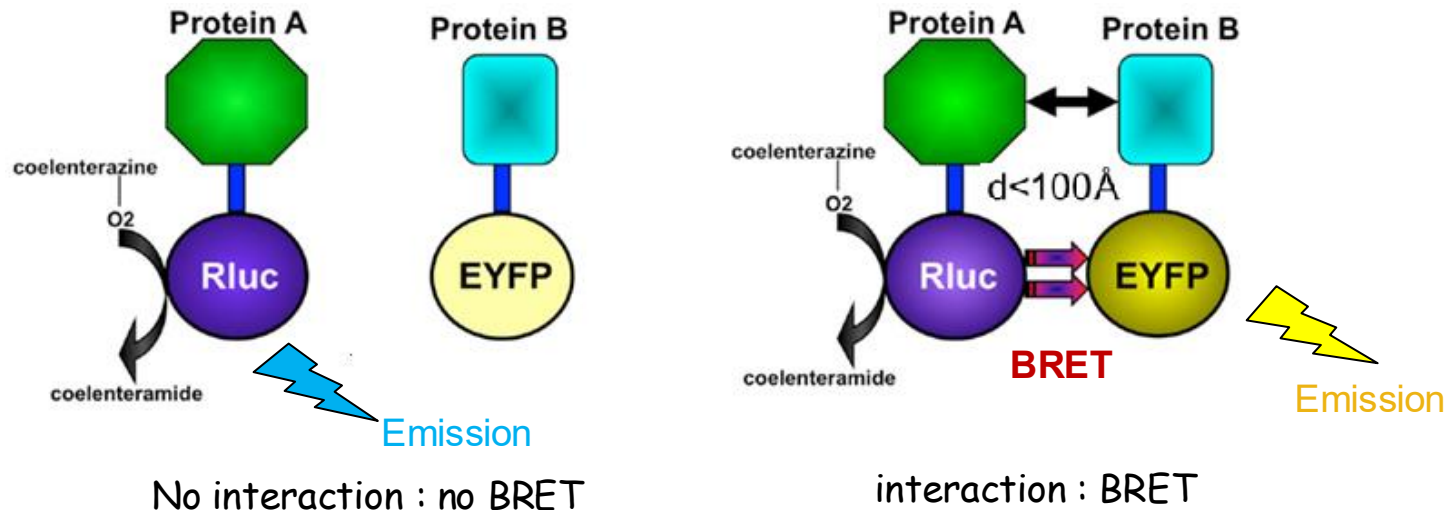


E. Analysis of protein interactions using FRET/BRET imaging

FRET : Fluorescence resonance energy transfer



BRET: Bioluminescence resonance energy tranfert



Protein A is fuse to a bioluminescent enzyme: luciferase (Rluc, from the coral *Renilla reniformis*)

E. Analysis of protein interactions using FRET/BRET imaging

Example of β -Arrestin-2 interaction with CXCR4 (GPCR) upon ligand binding

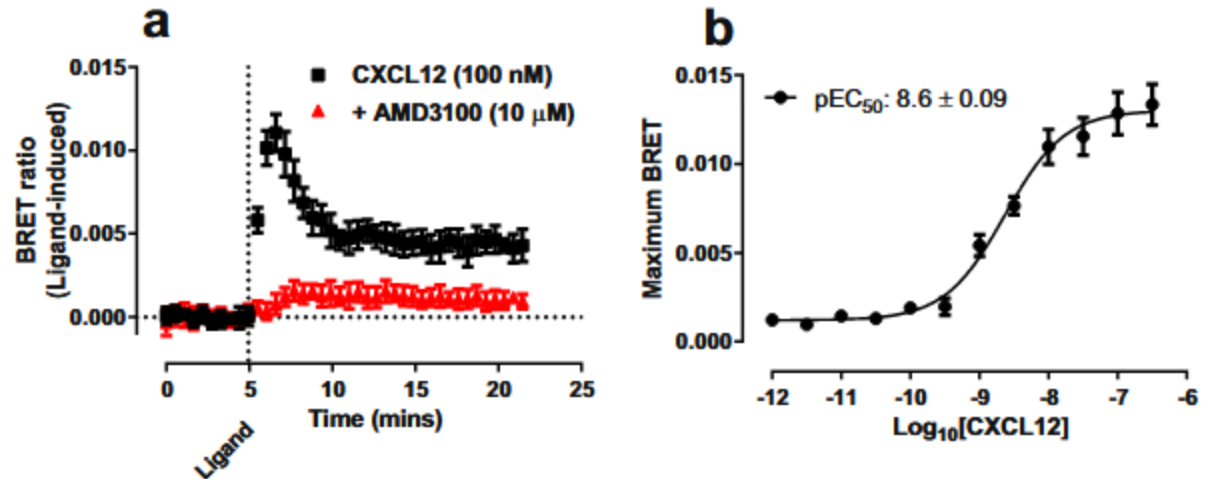
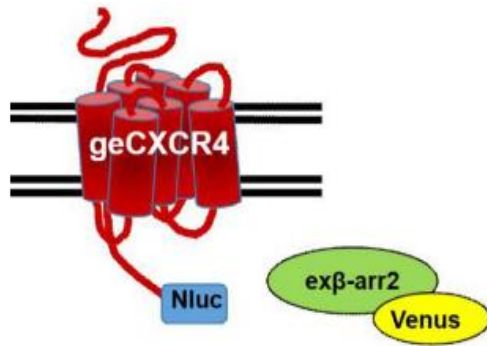


Figure 1. Monitoring β -arrestin2 recruitment to CXCR4/Nluc using BRET. HEK293FT cells expressing CXCR4 fused to Nluc (geCXCR4/Nluc) transiently transfected with cDNA coding for β -arrestin2/Venus (ex β -arr2/Venus) were used to (a) determine ligand-dependent CXCL12 (100 nM) recruitment of β -arrestin2 to CXCR4 in the absence or presence of the CXCR4 antagonist AMD3100 (10 μ M). (b) Application of CXCL12 (1 pM–300 nM) resulted in a concentration-dependent recruitment of ex β -arr2/Venus to geCXCR4/Nluc.

Q1: What is the BRET ratio ?

Q2: Analyze the figures. How is the curve in B obtained ?

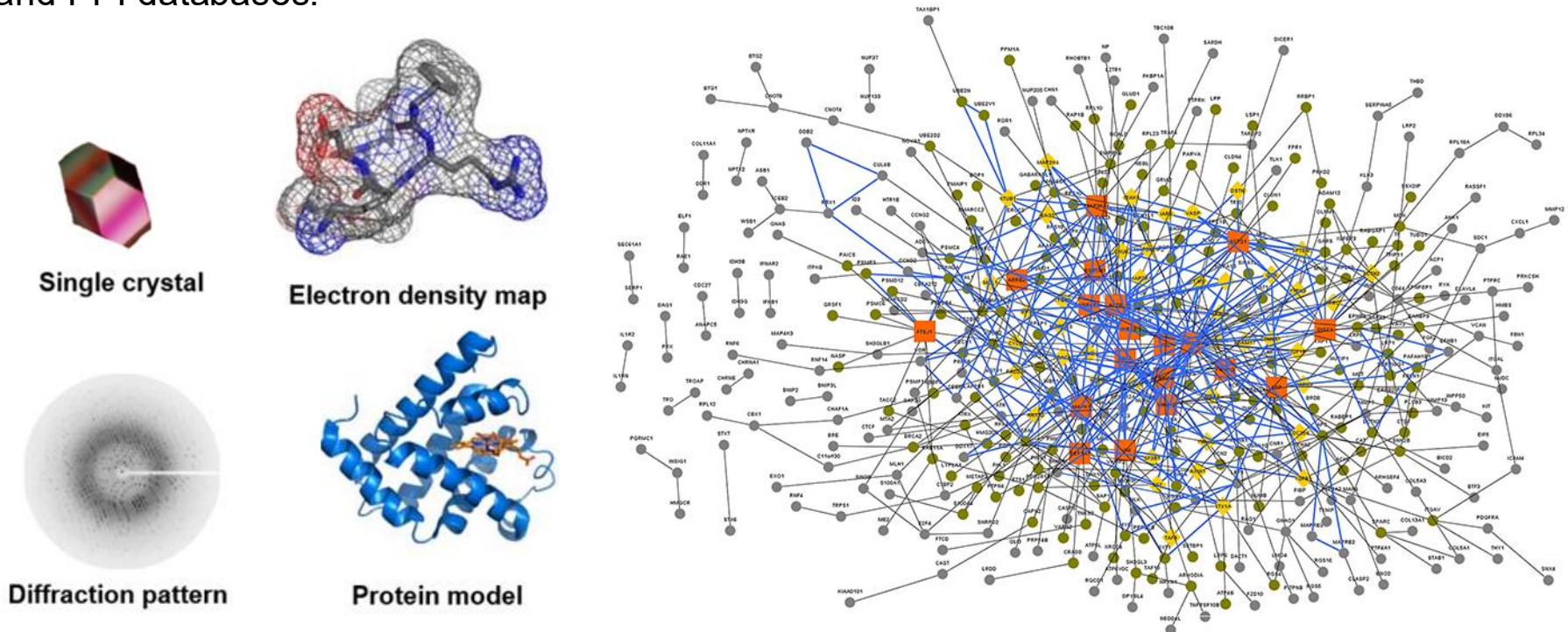
F. Conclusions

The study of PPIs is essential for the understanding of protein function in an healthy or diseased organism.

Many methods to investigate PPIs exist and continuously evolve to be improved.

Structures of proteins and protein complexes, studied using X-ray crystallography is essential to understand PPI and search for new drugs and treatments.

Besides, large scale predictive PPI networks can be established using computational methods and PPI databases.



Annexes

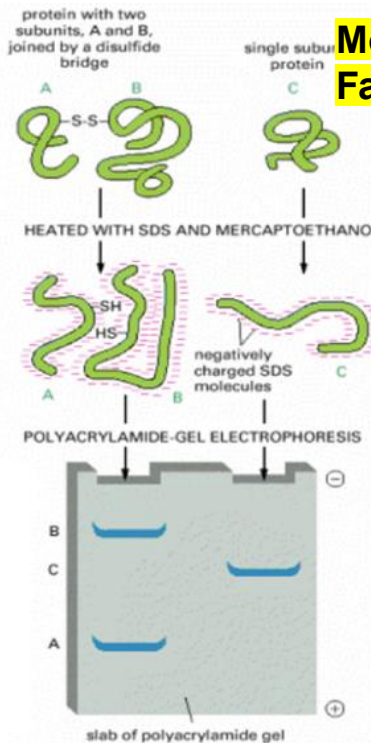
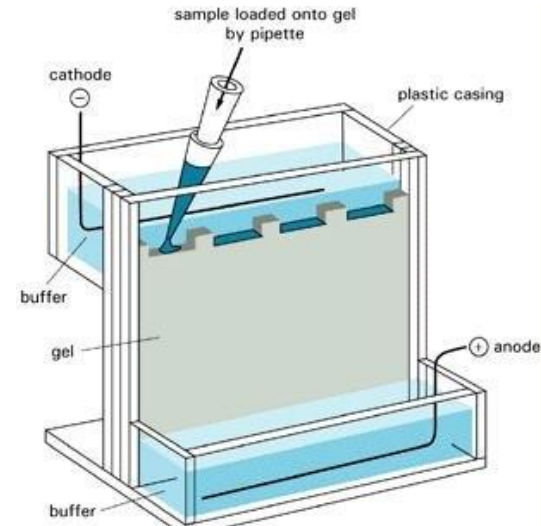
Western blotting (Immunoblotting)

See supplementary documents on moodle

Principe : Detect a protein through its interaction with a specific antibody. Allows a qualitative to semi-quantitative analysis. (Presence or absence, but also level of expression)

Step1: A-Separation of proteins by electrophoresis (SDS-PAGE)

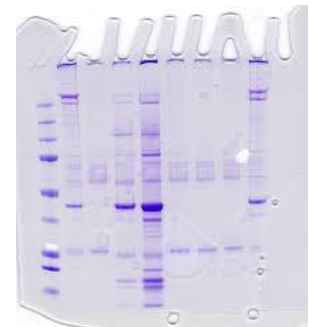
B-Transfer of proteins from gel to a membrane (nylon, nitrocellulose)



Modifier cette dia avec un schéma plus progressif de la méthode.
Faire apparaître les différentes possibilités après un SDS-PAGE.

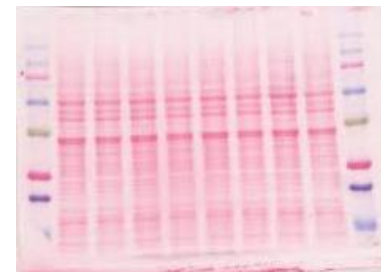


Protein visualization in polyacrylamide gel (Coomassie blue staining)



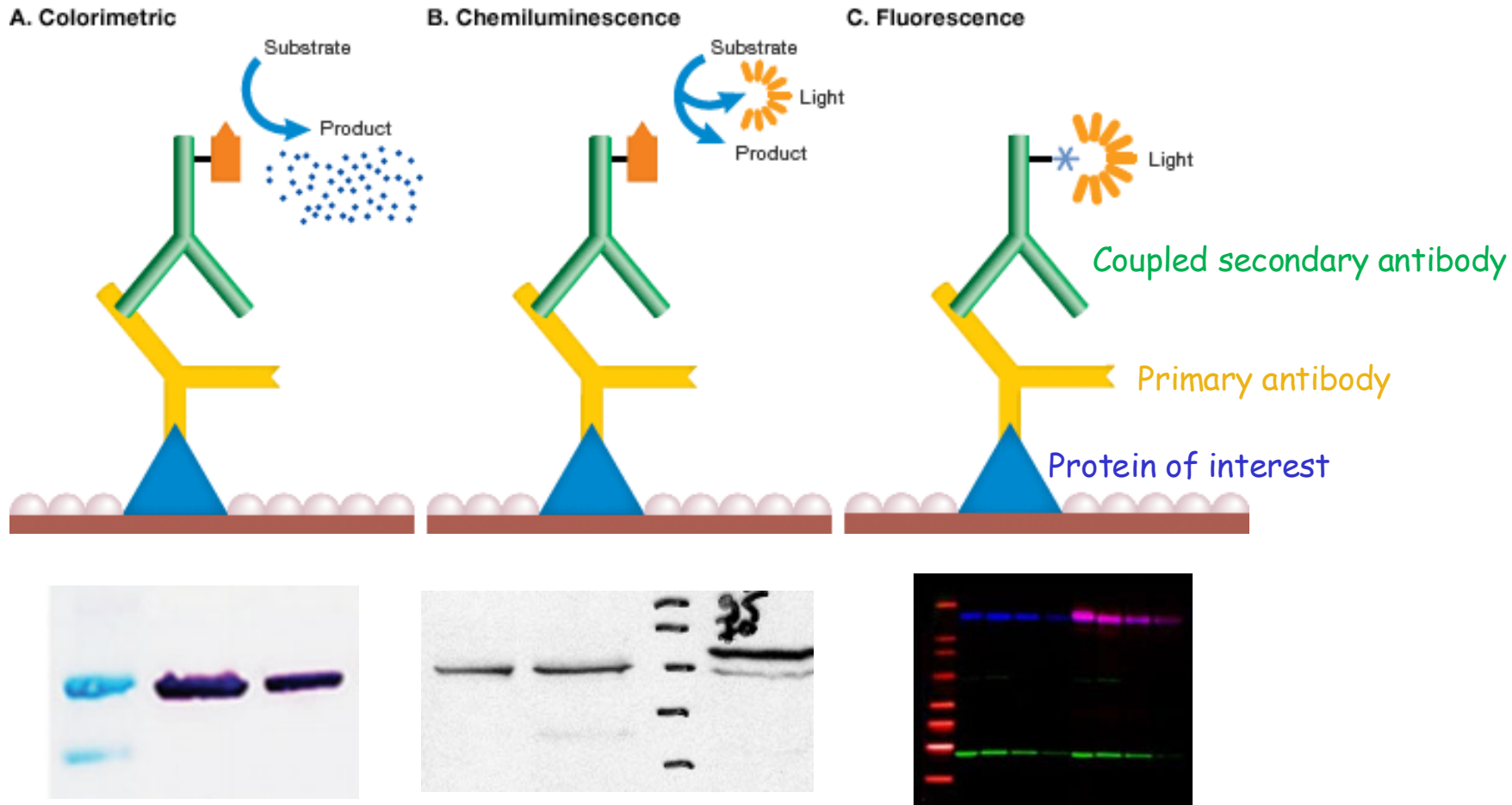
OR

Transfert proteins onto nitrocellulose or nylon membranes (Ponceau staining)



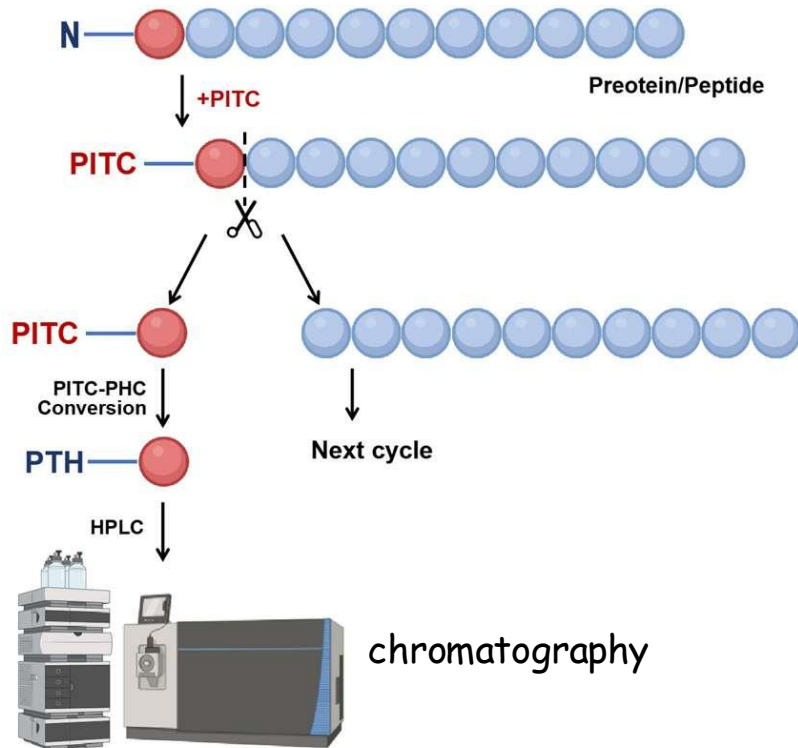
Western-Blotting (Immunoblotting)

Step 2: Membrane incubation with a specific antibody: specific detection of the protein of interest by interaction with an antibody coupled to a revelation system



A. Analysis of protein interactions using co-immunoprecipitation

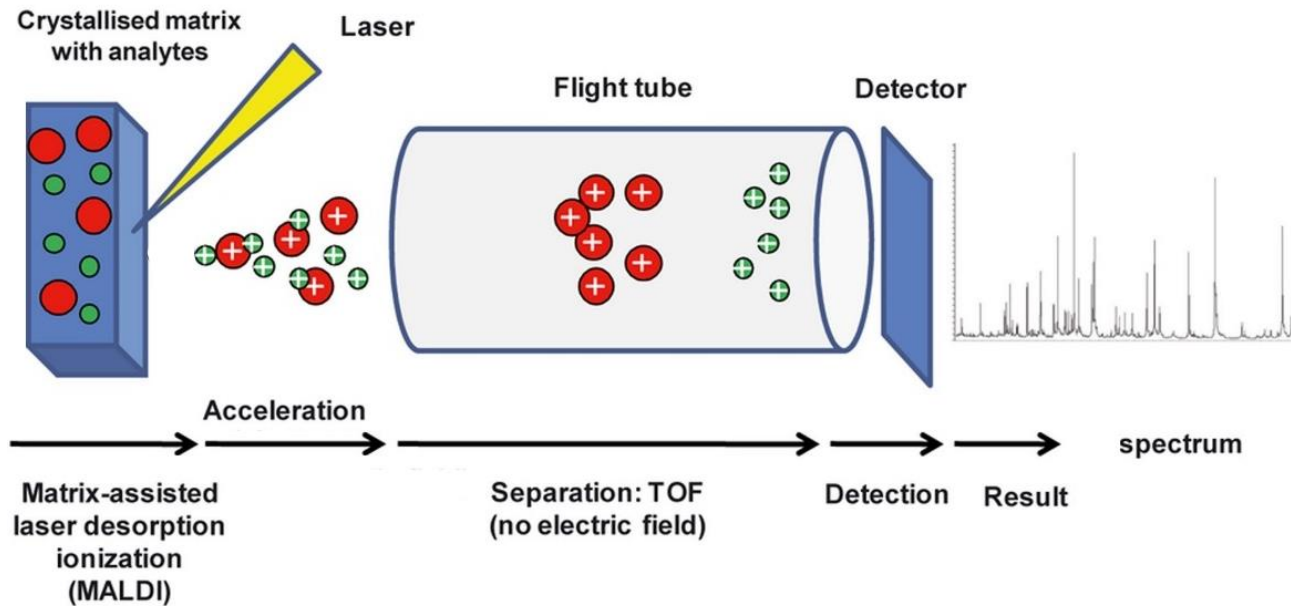
1) Sequencing of the N-terminus of the protein: the Edman's method



- After several cycles: determination of the N-terminal sequence (~20-30 amino acids)
- Identification of the protein by computer comparison of the obtained sequence with those of established databases

A. Analysis of protein interactions using co-immunoprecipitation

2) Sequencing by MALDI-TOF (Matrix Assisted Laser Ionisation Desorption – Time Of Flight)



- Flight time proportional to the mass of the peptide
- The mass of the peptide depends on its amino acid composition

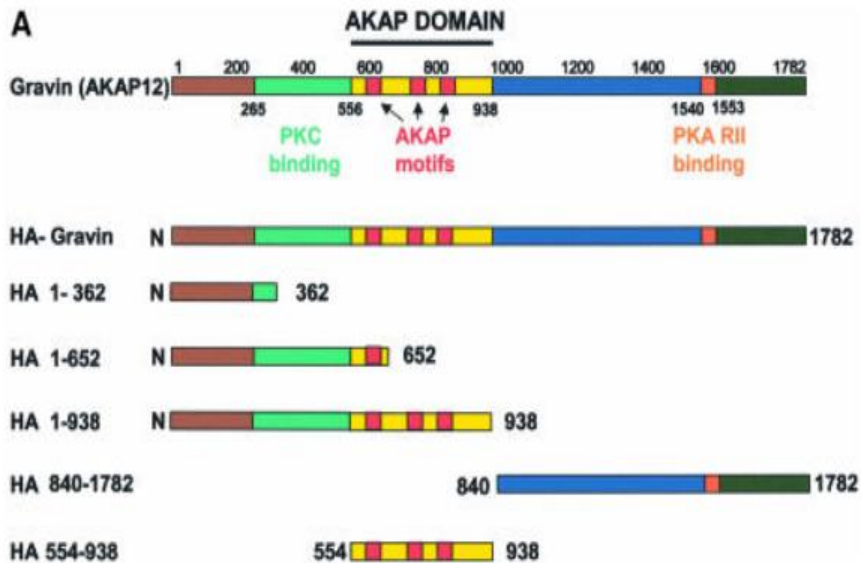
Comparison of the obtained peptides with those generated virtually from the protein sequences available in the databases

Annexe 2. Exercice:

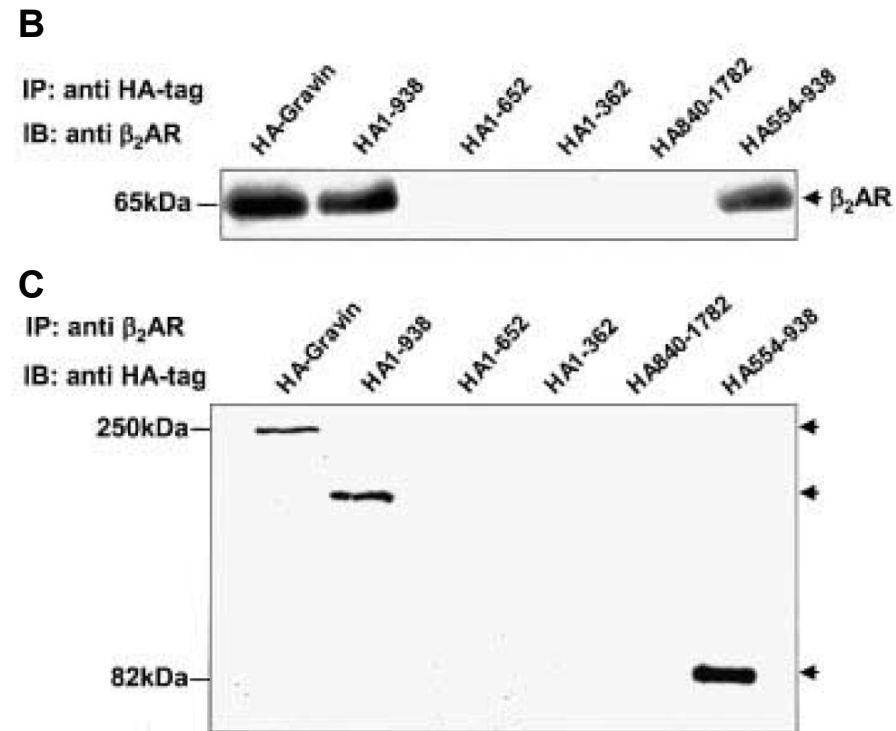
Analysis of protein interactions using co-immunoprecipitation

Example: Interaction of Gravin with the β_2 -adrenergic receptor

- Authors have generated plasmids for the expression of HA-Gravin fusion proteins. They cloned the full or truncated Gravin cDNA and added a N-terminal HA tag (A).
- The constructs were expressed in cells expressing the β_2 AR. After Lysis, authors performed co-IP using either an anti HA antibody (B) or a anti β_2 AR antibody (C).



Tao, Embo J 2003



Q1: Would it be possible to do an IP using an anti gravin antibody? Why did the authors add a HA-tag ?

Q2 Analyze figures (goal, method, results and conclusions).

Q3: In case antibodies for the HA tag were not available, what other technique(s) can be used to address the authors' question ? Propose a short protocol.